IRA A. FULTON SCHOOLS OF ENGINEERING

Leading engineering discovery and innovative education for global impact on quality of life.

EEE 335 Course Introduction

Lecture 0 - Introduction

1/13/2026

James McDonald

ARIZONA STATE UNIVERSITY

Who's this dude?

- James McDonald– Professor, Mr. McDonald, James, "Something Nice"
- Over 37 years in industry – Motorola, BBN, Medtronic, Cactus Semiconductor
- Co-founded fabless semiconductor company "Cactus Semiconductor"
  - President / CEO for 16 years
  - Focused on custom integrated circuits (ASICs) – Medical, Portable
  - Sold company to Cirtec Medical (2018) retired in February 2020.
- My goal is to teach and help facilitate learning. I don't do research.

Instructor Information

James McDonald, Professor of Practice, ECEE Department

Office Hours:

Tuesday	1:30 pm – 3:30 pm
Friday	10:30 am – 12:30 pm

Also available by appointment: Send email with Subject: “Appointment Request”.
I am available to help you. Take advantage if you are having difficulties.

Office Location:
GWC 438

Contact Information
Phone: (602) 543-6701

Zoom: https://asu.zoom.us/j/5732009374

Email: jtmcdona@asu.edu

EEE 335 Course

Title: Analog and Digital Circuits
Units: 4
Lecture Location/Dates/Times: SCOB 250, T/TH 12:00 PM – 1:15 PM
Course Website: Canvas in myASU

Course Objectives:

- MOSFET Operations (Review of EEE 334)
- Analyze basic building blocks of digital microelectronics circuits
  - CMOS Inverter and CMOS Logic
  - Pass Transistors
  - Latches, Flip-Flops & Memory Circuits
- Analyze basic building blocks of analog microelectronics circuits
  - Single stage single-sided amplifiers
  - Frequency response of amplifiers
  - Cascode & Differential Amplifiers
- Apply the methods learned in class to implement and simulate CMOS analog and digital circuits. (Cadence IC Schematic Capture & Simulation)

Prerequisites

- **EEE 334**
  - DC and small signal AC analysis of semiconductor circuits.
  - MOSFET and junction diode models.
  - Understanding of ideal op-amp operation.
  - Inverters and basic CMOS gates.

Related Courses

- Basic Circuits
  - EEE 202 Circuits I
  - EEE 334 Circuits II
  - EEE 335 Analog and Digital Circuits
- Digital Circuits Path
  - EEE 425 Digital Systems and Circuits
  - EEE 525 VLSI Design
  - EEE 625 Advanced VLSI Design
- Analog Circuits Path
  - EEE 433 Analog Integrated Circuits
  - EEE 523 Advanced Analog Integrated Circuits

EEE 335 is an area pathway course for Electronic & Mixed Signal Design emphasis.

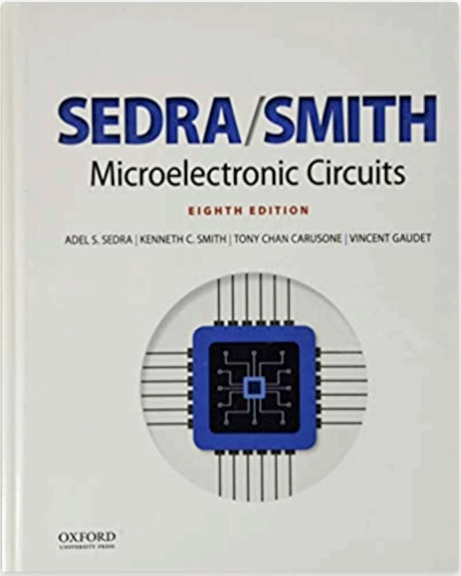
Course Material

In this course, you will learn through:

- Book, Lectures, and Supplemental Reading
- Homework Assignments/Quizzes
- Laboratory Assignments
- Exams

BOOK:

A.S. Sedra and K.C. Smith,
Microelectronic Circuits, 8th Edition,
Oxford University Press.



See your textbook reading agenda
in your Canvas “Welcome” module

EEE 335 Course Topics

- MOSFET I-V Model
  - Large-signal model (Bias)
  - Small-signal model
- Digital Circuits
  - Inverter
  - Logic gates
  - Memory circuits
- Analog Circuits
  - Basic CS, CG, CD amplifiers
  - Frequency response analysis
  - Cascode & Differential amplifiers

Course Grading

• 8 Homework problem sets:	15 points each	(15%)	Lowest Dropped
• 5 Laboratories:	25 - 50 points each	(25%)	
• 5 Canvas Quizzes:	10 points x 4, 20 points x 1	(10%)	Take 2x, High Score
• 1 Midterm Exam:	100 points	(25%)	Digital Portion
• Final Exam:	100 points	(25%)	Analog Portion

Total: 100%

- **SRAM Simulation Bonus (added at end) 1.0%**
- **Course Evaluation Bonus (added at end) 0.5%**
- **3P's Push-Pull-Points (added at end) 0.5%**
  - Attendance
  - Assignment Submission (on-time)
  - Takes advantage of additional extra credit
  - Quality of work submitted
  - Participation in class
  - Used in borderline grade situations – typically 0.0%

Grades

- Plus and minus grades are given (A+, A, A-, B+,B,B-,C+, C, D, E) –> See syllabus
- Do NOT generally plan to grade exams on a curve (subject to change)
- If extra credit and/or second opportunity to earn points back, take advantage of it!

Lectures

- I will use Canvas to post some lecture notes and communicate with the class.
- I will try to post lecture **slides** prior to class (Take notes directly on those slides.)
- Lecture recordings, worked in-class problems and solutions will typically be posted within 24 hours after the class.
- Lectures will be a combination of a PowerPoint presentations and hand derivations/examples of problems.
- Lecture numbers are based on topics not a class period. Multiple lectures may be covered in one class period. (e.g. Class 4 covers lectures 3 & 4). This is lecture 0.
- Attendance (in-person) is expected. I will periodically take attendance. This will be used as part of a “3P’s” bonus. This bonus can help (boost) your grade if you are borderline between 2 grades.
- In addition, participation in class, attending office hours and not being disruptive and participating in class can help your “3P’s” bonus. Initial bonus score for all = 0%

Homework Assignments

- I will use Canvas to post HW assignments, solutions, updates, and supplemental material.
- Homework assignments will be submitted through Canvas on the specified due date & time.
- There are 8 HW assignments. Each Homework is worth 15 points (15% of grade)
- **Late HW assignments will be penalized 50%. Homework more than 3 days late will receive a zero.**
- I also drop the lowest homework score (This can be used during a really busy week.)
- Recommend doing all homework to prepare you for exams.
- Homework #8 has a bonus problem.

Laboratory

- 5 Laboratory Projects.
- Hand analysis and simulation of circuits.
- Requires CAD software (Cadence) for simulation.
- CAD will be access remotely through the ASU servers or directly in GWC295. More information on this will follow. You will have instructions for accessing the CAD software in your “Lab Remote Access” module on Canvas.
- Students will electronically submit lab reports through Canvas.
- Laboratory Projects = 25% of your total grade.

This semester, you may complete your labs online using remote access to Apporto, or you may complete your labs in the CAD lab room GWC 295.

Your lab TAs will answer questions online via Ed Discussion, through emails, or in-person if you attend the GWC 295 hours during your TAs’ times.

GWC 295 Open: Mon-Thurs 9am to 9pm; Fri 9am-6pm
Your TA in-person times will be posted soon.

The in-person lab situation may evolve through the semester, so please keep updated on lab opening times and such.

Laboratory Assignments

- *Lab 1: Cadence Tutorial*
- *Lab 2: Inverter Simulation*
- *Lab 3: Design and Analysis of CMOS gates*
- *Lab 4: Design and Analysis of a Common Source Amplifier with Active Load*
  - *Lab #4 is separated into 2 parts. Due date is after second part.*
- *Lab 5: Design and Analysis of a Differential Amplifier with Passive/Active Loads*

Labs are an individual assignment.

Late lab submission will have 20% deducted per day for up to 3 days. Zero on the lab after 3 days.

Laboratory TAs

- Lab TAs are available via Ed Discussion to help you. They will answer Ed Discussion and email questions, and setup zoom calls with you as needed.
- They do NOT grade your laboratory reports. Please email lab-related grading questions to instructor or TA. They will work with the ***Grader*** to answer questions.
- TAs do NOT have responsibility for homework assignments, only labs.
- TA names are posted in your Canvas syllabus. ZOOM and in-person help hours will be posted by next week.

TA Names and Hours

- UGLAs & TA for EEE 335 Labs and Class work (TA)
  - TBD1 UGLA
  - TBD2 UGLA
  - TBD3 UGLA
  - TBD4 UGLA

- Office / Labs hours to be posted on Canvas

EEE 335 Quizzes

- 5 Quizzes in total worth 10 points each (#1-4) and 20 points Quiz #5
- Quizzes are open book, open notes and a calculator may be used.
- No AI or internet allowed.
- Quizzes are not timed
- Two (2) attempts per quiz with the highest grade accepted
- All correct answers will be shown 3 days after the due date.
  - Only your correct answers should show after submitting quiz.
- No Late quizzes accepted

EEE 335 Exams

- Midterm Exam and a Final Exam

- Exams are closed book, but 'formula sheets' may be allowed (details explained later).

- Final Exam time is scheduled by the University.
  - Date: Tuesday, May 5, 2026
  - Time: 12:10pm – 2:00pm
  - Room: SCOB 250

- Each exam is worth 25% of your grade

  Midterm 25% (Focused on 1st half of course – MOSFET & Digital)
  Final 25% (Focused on 2nd half of course – Analog)

Ed discussion

- Please ensure that you register with Ed Discussion, if you do not see this in your Canvas user account.
- TAs & students will use Ed Discussion in this course to communicate questions.
- Often times, you will find answers to your questions in others' responses.
- TAs will answer questions on Ed Discussion. If you do not understand or do not get a complete answer contact the TA (labs) or you can email me after you've explored these options first.
- When you submit a question, please use a category tabs related to the question (e.g. HW 1, Lab 1, etc.) or if emailing use the "Subject line". (Lab 1 Question)
- Don't ask the instructor questions on Ed Discussion. Email the instructor.

Additional Information

- Refer to Canvas for course material.

- Grader will be grading your homework and lab assignments:
  - Email me for grading inquiries on homework.
  - Email your TA for grading inquiries on lab and copy me on the email.

- Most notes will be a combination of slides and hand analysis during lecture. The notes will follow content within the book. It is required that you have access to the course book, since lecture notes will not cover all material.

- 2-3 copies of our textbook can be used in the Noble Library. (Based on prior semesters)

Notes on Policy

- Final Exam: Only those with more than 3 exams on the same day may change his/her final exam date.

- There are no make-up tests. If you plan to miss a test due to excused absence, contact the instructor immediately and provide an official medical excusal for this absence. If you miss a test due to a medical emergency, the instructor requires proof of hospitalization/other.

- For any questions on graded HW and Exams, you must communicate with the instructor. DO NOT CONTACT the TAs for grading problems unrelated to laboratory assignments.

- Questions regarding homework/lab/exam grading must be asked **within one week** after the homework/lab/exam is returned.

- Aid, which results in a student submitting work that does not reflect the student's understanding and effort, is a violation of ASU's academic integrity policy. No AI.

For Next Class / Week

Browse through your textbook to become familiar with the flow and chapter topics.

Read your course **syllabus**, **"Welcome"** and **"Lab Remote Access"** module.

READ:

MOSFET Operation: Chapter 5 (See Unit 1: Overview Module)

- Device model
- Regions of operation
- DC analysis

Laboratory: If remote access is desired, begin looking at setting up your remote access per Canvas instructions.